poly-maker backtest

Tape replay of the configured strategy over the last 45 days. Generated 2026-10-27 21:16 UTC. Config: /Volumes/SHARGE-Data/polymarket-bot/poly-maker/config/. Three scenarios per market: **engine as-is** replicates the live engine's known defects; **fixed** has them corrected; **fixed + defenses** adds the inventory-lean floor and the bounded taker trade-out from the configured profile. Rewards are an estimate at an assumed 1% share of each market's daily pool.

Headline

Market	Scenario	Fills	Fill $	Trading PnL	Markout 5m	Max DD	Scored	Rew @1%	Total
Will Gavin Newsom win the	engine as-is	38	92	-1.40	+0.45	2.85	9%	+1.34	-0.05
Will Gavin Newsom win the	fixed	55	279	-0.36	+0.16	1.03	90%	+14.17	+13.81
Will Gavin Newsom win the	fixed + defenses	45	239	-0.24	+0.21	0.96	90%	+14.17	+13.92

Verdict

Will Gavin Newsom win the 2028 Democratic presidential nomin: trading PnL -0.36 fixed vs -1.40 as-is over 45 days, -0.24 with defenses (0 trade-outs, max inventory $36 vs $38). With rewards at a 1% pool share the best total is +13.92 vs -0.05 as-is. Size-weighted 5-minute markout +0.5 ticks; quotes scored rewards 90% of sampled minutes (as-is 9%).

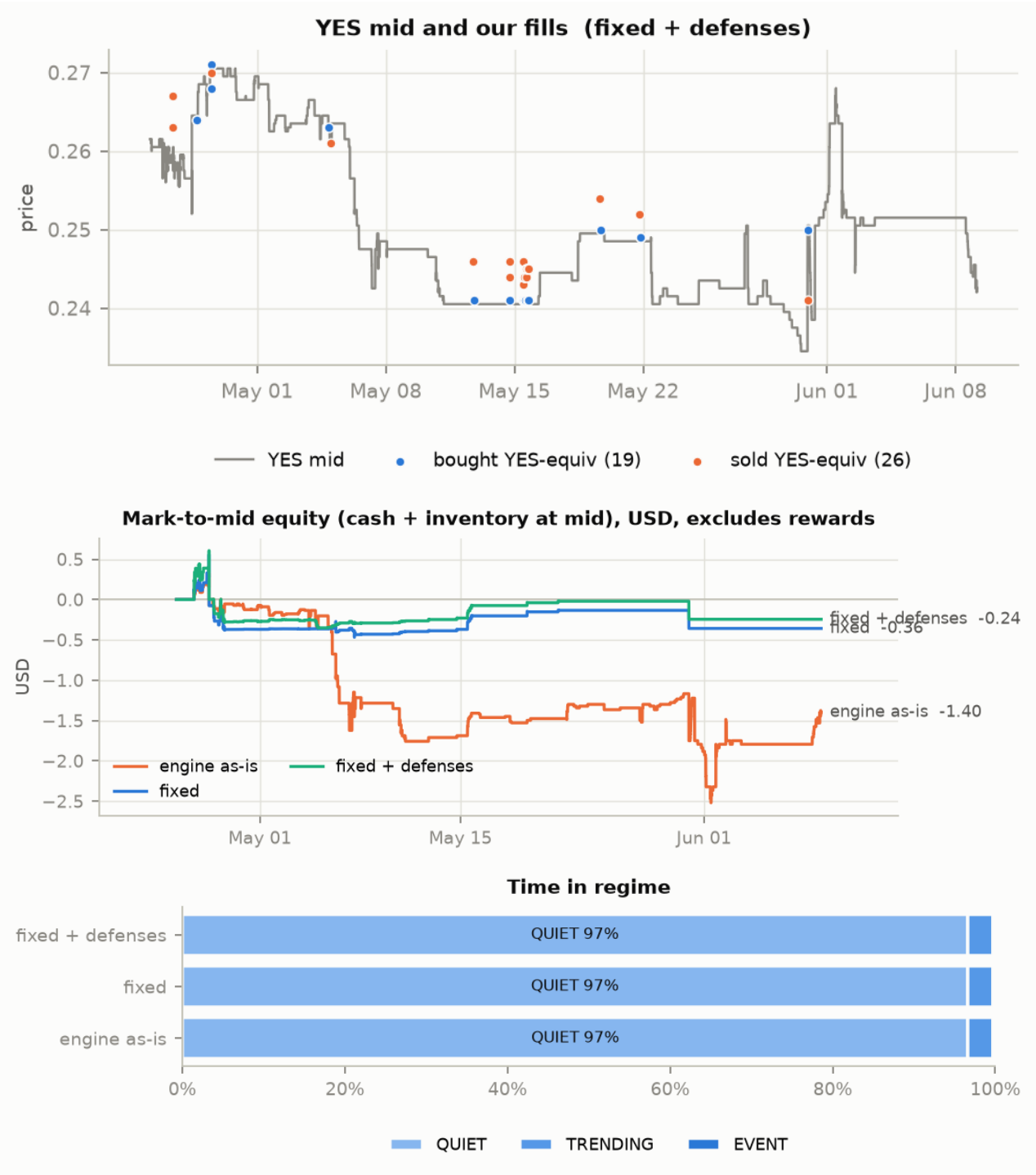
Reading the table: **Trading PnL** is cash from fills plus inventory marked at the final mid, so it includes unrealized inventory risk. **Markout 5m** sums, over all fills, how far the mid moved for or against us five minutes after each fill; a negative number is adverse selection paid. **Scored** is the share of sampled minutes in which our resting quotes met the reward program's size and band rules on both sides (Qmin > 0). **Rewards** = daily pool x days x scored fraction x assumed share; see the sensitivity table on each market page.

Method and assumptions

- **Strategy code under test is the repo's own pure core:** compute_fair_value, Vol/Flow/Markout estimators, RegimeMachine, construct_quotes, reconcile, RiskManager over a StateStore. The engine's I/O is replaced by the replay harness; nothing in strategy/ was modified for the backtest.
- **Data:** taker trade tape from poly_data on-chain fills (orderFilled.csv) and the CLOB /prices-history minute series for the YES token. Trades on the NO token are mapped to YES space (taker BUY NO at q = taker SELL YES at 1 - q). The on-chain tape holds every fill since V2 genesis, one print per taker order on the asset it traded (maker legs of the same match are not double counted).
- **Synthetic book:** no historical L2 exists, so best bid/ask = minute mid -/+ half the assumed spread (parameter spread_ticks) with a constant touch depth. Microprice = mid. Sub-minute book jitter, dust at the touch, and air pockets are therefore NOT modeled; the live regime machine will fire more often than this replay shows.
- **Fills (volume-ahead queue):** an order that joins or sits behind the touch starts with touch_depth shares ahead of it; prints at its level drain that queue before we fill; prints through the level fill us fully. Our presence is assumed not to change the tape (standard maker-backtest assumption; optimistic on thin markets).
- **Mint/merge matching is honored:** a taker SELL YES fills our BUY YES and our SELL NO exits; a taker BUY YES fills our SELL YES and our BUY NO. Pairs are marked at 1.00 total (merge not simulated; capital lock-up is reported, not penalized).
- **Rewards:** each minute we compute Q = S x ((v - s)/v)^2 per in-band order of at least the reward min size, Qmin = max(min(Qb, Qa/3), min(Qb/3, Qa)) for mids in [0.10, 0.90]. Competition is unobservable, so earned rewards = pool x scored fraction x an assumed share; three shares are shown per market.
- **Fees/rebates:** trade-outs pay the market's taker fee (rate x p x (1-p) per share); maker fills pay nothing and, on fee-enabled markets, are credited rebate_rate x the taker's fee (0 on fee-free markets such as politics).
- **Scenario 'engine as-is'** replicates: (1) markout recorded in token space against YES fair value, (2) raw aggressor sign on NO-token prints feeding the flow EWMA, (3) exit urgency never ramped. **'fixed'** corrects all three and runs the strategy core as it was before the defenses (every defense knob off). **'fixed + defenses'** runs the configured profile: inventory-lean floor, exits priced off the leaned reservation price, EVENT keeps exits alive, inventory measured as capital at risk, sub-minimum remnants replaced, and the bounded taker trade-out.
- **Toxicity memory:** an earlier candidate fix let adverse-selection toxicity age out over 30 minutes. Measured on both markets it re-engaged full size after every bad fill and roughly quadrupled the trading loss, so the shipped default keeps a long memory (no decay); the sensitivity table shows a 6-hour half-life for reference.
- **Risk:** the config's [risk] caps and daily-loss kill are active, reset at each UTC day boundary. Neg-risk group cost is zero (single markets).
- **Not modeled:** WS staleness, heartbeat gaps, post-only rejections, partial batch failures, data-API lag, gas or relayer costs, and competition reacting to our quotes.

Will Gavin Newsom win the 2028 Democratic presidential nomination?

slug will-gavin-newsom-win-the-2028-democratic-presidential-nomination-568 - profile **fifty** - tick 0.001 - reward min size 50 sh, band 5.5c, pool $35/day - taker fee 0.0%, rebate 0%
Window 2026-04-25 to 2026-06-09 (45.0 days): 6,936 taker prints, $260,835 tape notional, 64,587 mid samples, YES mid range 0.234 to 0.271.



Will Gavin Newsom win the 2028 Democratic presidential nomination? - detail



Metric	engine as-is	fixed	fixed + defenses
Fills (count / shares / USD)	38 / 243 / 92	55 / 511 / 279	45 / 391 / 239
Fill rate vs tape (fills per 100 prints)	0.5	0.8	0.6
Trading PnL (cash + inventory at mid)	-1.40	-0.36	-0.24
of which cash	-38.25	-0.36	-0.25
of which inventory value	+36.85	+0.00	+0.01
Final position YES / NO (shares)	4 / 48	0 / 0	0 / 0
Max net inventory (shares) / max inventory USD	68 / 43	51 / 38	48 / 36
Max drawdown (mark-to-mid)	2.85	1.03	0.96
Daily PnL Sharpe (annualized, no rewards)	-3.14	-1.87	-1.08
Markout 1m: size-wtd ticks / total USD / adverse share	+2.19 / +0.53 / 3%	+0.72 / +0.37 / 47%	+0.76 / +0.30 / 38%
Markout 5m: size-wtd ticks / total USD / adverse share	+1.87 / +0.45 / 3%	+0.32 / +0.16 / 49%	+0.54 / +0.21 / 38%
Markout 30m: size-wtd ticks / total USD / adverse share	+1.80 / +0.44 / 3%	+0.25 / +0.13 / 49%	+0.46 / +0.18 / 38%
Trade-outs (count / shares / taker fees USD)	0 / 0 / 0.00	0 / 0 / 0.00	0 / 0 / 0.00
Maker rebates credited (USD, inside PnL)	0.00	0.00	0.00
Orders placed / cancelled	5,586 / 5,583	8,309 / 8,294	8,498 / 8,484
Avg order lifetime (min)	31.3	15.6	15.2
Minutes scoring rewards (Qmin > 0)	5,510 / 64,587 (9%)	58,120 / 64,587 (90%)	58,107 / 64,587 (90%)
Mean Qmin while scoring (shares)	22	12	12
Est. rewards at 5% pool share	+6.72	+70.86	+70.85
Est. rewards at 10% pool share	+13.44	+141.73	+141.70
Est. rewards at 20% pool share	+26.87	+283.46	+283.40
Total PnL incl. rewards @1%	-0.05	+13.81	+13.92
Minutes halted (kill switch / caps)	0	0	0

Engine defects: measured cost

Two steps per market: the engine bug fixes (fixed minus as-is) and the defenses on top (defended minus fixed). Positive PnL/markout = the step helps; negative max-inventory and drawdown = less risk carried. The replay sees only minute mids and a synthetic book, so treat magnitudes as directional.

Market	Step	Trading PnL	Markout 5m	Scored min	Fills	Max inv $	Max DD
Will Gavin Newsom win the 2028	bug fixes (fixed - as-is)	+1.04	-0.29	+52,610	+17	-6	-1.82
Will Gavin Newsom win the 2028	defenses (defended - fixed)	+0.11	+0.04	-13	-10	-2	-0.07

Sensitivity: one knob at a time on top of 'fixed + defenses'

Each row changes exactly one profile value. Total = trading PnL + rewards at the assumed 1% share. One month of one market is not enough to adopt a tuning from this table; it shows which levers matter.
Will Gavin Newsom win the 2028 Democratic presidential nomination?

Variant	Fills	Trading PnL	Markout 5m	Scored	Max DD	Trade-outs	Total @1%
configured (defended)	45	-0.24	+0.21	90%	0.96	0	+13.92
c_tox x2	45	-0.14	+0.37	70%	0.95	0	+10.95
skew floor 4 ticks	46	-0.30	+0.13	90%	0.75	0	+13.90
trade-out drift 5	54	-0.32	+0.43	91%	0.99	2	+13.99
trade-out drift 20	45	-0.24	+0.21	90%	0.96	0	+13.92
no trade-out	45	-0.24	+0.21	90%	0.96	0	+13.92
tox decay 6h	77	-1.27	-0.85	90%	1.88	0	+12.92
entry yields to exit	45	-0.24	+0.21	90%	0.96	0	+13.92

Caveats before acting on this

- A trades-only replay cannot see queue competition, dust at the touch, pulled liquidity, or gaps. On thin markets the live engine will be picked off more than shown here.
- The public tape under-reports flow (25-40% of Gamma's 24h volume on the run date), so expect more fills live than in this replay, in both directions.
- Rewards dominate the economics of both configured markets. The share assumption is the single largest uncertainty; measure it live by comparing paid rewards to the scored fraction over a few days.
- Window is one month of recent history. Re-run with --days and other markets before generalizing; the data cache makes re-runs offline.
- The trade-out is modeled as an immediate fill at the synthetic best bid for the capped size. Live, a FAK can fill partially or not at all when the bid has moved; the engine then waits out the cooldown rather than chasing.
- Remaining review findings not addressed by any scenario: the flow z-score bound (flow TRENDING trigger unreachable), the consecutive-recompute jump detector, own orders polluting the microprice, and the never-improve-the-touch placement rule.